

ZHONGMOU CHAO

Eric and Wendy Schmidt AI in Science Postdoc Fellow

Cornell University

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EDUCATION

Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA

Doctor of Philosophy in Chemical Engineering 12/2020

Department of Chemical Engineering, Columbia University, NY, USA

Master of Science in Chemical Engineering 02/2015

College of Chemistry and Chemical Engineering, Nanjing Tech University, China

Bachelor of Engineering in Chemical Engineering 07/2013

RESEARCH EXPERIENCE

Smith School of Chemical and Biomolecular Engineering, Cornell University, NY, USA

Eric and Wendy Schmidt AI in Science Postdoc Fellow, [Daniel Lab](#) and [Frazier Lab](#) 09/2023 – Present

- Developed an machine learning (ML) model for protein function prediction: better protein function prediction by modeling survivorship bias
- Leveraged Evo-PU to predict influenza viral protein functions and screen variants of concern
- Integrate ML, cell-free protein synthesis and bioelectronics to design drug delivery systems leveraging the functionalities of *de novo* viral proteins

Postdoc Associate, [Daniel Lab](#) 05/2021 – 08/2023

- Pioneered a bioelectronic sensor built on supported lipid bilayer (SLB)
- Developed viral infection-on-chip platform leveraging SLB-on-bioelectronics
- Integrated cell-free protein synthesis with bioelectronics for protein function studies

Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA

Graduate Research Assistant, [Fullerton Lab](#) 10/2015 – 04/2021

Dissertation Title: *Silver Nanofilament Formation/Dissolution Dynamics Through a Polymer/Ionic liquid Composite by Direct-write*

- Investigated the impact of an ionic liquid on silver nanofilament formation kinetics through a solid polymer electrolyte
- Identified unique filament growth dynamics resulting from a competition between electrical double layer screening by the ionic liquid and silver electrodeposition
- Demonstrated how this newly identified growth mechanism could be useful for application in flexible electronics and neuromorphic computing

Department of Chemical Engineering, Columbia University, NY, USA

Graduate Research Assistant, [West Lab](#) 01/2014 - 06/2015

- Non-Photosynthetic Production of Biofuels from CO₂
- Production of Biofuels from Sulfur by A. Ferrooxidans Cells

PUBLICATIONS

1. **Chao, Z.**, Buathong, P., Selivanovitch, E., Daniel, S., and Frazier, P.. "Better Protein Function Prediction by Modeling Survivorship Bias." **Submitted to NeurIPS 2026**
2. Hattar, A., Alhammadi, J., Treiber, J., Hoven, D., **Chao, Z.**, Offenhaeusser, A., Daniel, S., Owens, R., Salleo, A., Santoro, F. and Pappa, A.M.. "Biomembranes in Hybrid Living Bioelectronics." *Nature Protocols*, **accepted (2026)**.
3. Treiber, J., **Chao, Z.**, McCoy, R., Lu, Z., Wheeler, A., Thiburce, Q., Daniel, S., Owens, R., Salleo, A.. "Reproducible High-Impedance Supported Lipid Bilayers on PEDOT:PSS." *Advanced Functional Materials*, e28703 **(2026)**. ([//doi.org/10.1002/adfm.202528703](https://doi.org/10.1002/adfm.202528703))
4. Lu, Z., Treiber, J., Kallitsis, K., Selivanovitch, E., Wheeler, A., Lopez-Cavestany, M., **Chao, Z.**, Barron, S.L., Park, J.A., Hoven, D., Scheeder, A., Hass, B.M., Kaminski, C.F., Salleo, A., Pappa, A.M., Daniel, S. and Owens, R.. "Automated Label-Free Assay for Viral Detection and Inhibitor Screening via Biomembrane-Functionalized Microelectrode Arrays." *Advanced Materials*, e01985 **(2025)**. ([//doi.org/10.1002/adma.202501985](https://doi.org/10.1002/adma.202501985))
5. Su, Z., **Chao, Z.**, Jiang, V., Daniel, S. and Banta, S.. "Site Directed Mutagenesis of the Cyc2 Outer Membrane Protein from Acidithiobacillus ferrooxidans Reveals a Critical Role for Bound Iron Atoms in Extracellular Electron Transfer." *Small*, 2408837 **(2025)**. ([//doi.org/10.1002/sml.202408837](https://doi.org/10.1002/sml.202408837))
6. **Chao, Z.**, Selivanovitch, E., Kallitsis, K., Lu, Z., Pachaury, A., Owens, R., and Daniel, S.. "Recreating the biological steps of viral infection on a cell-free bioelectronic platform to profile viral variants of concern." *Nature Communications*, 15, 5606 **(2024)**. ([//doi.org/10.1038/s41467-024-49415-6](https://doi.org/10.1038/s41467-024-49415-6))
7. Liu, S.S., White, J.M., **Chao, Z.**, Li, R., Wen, S., Garza, A., Tang, W., Ma, X., Chen, P., Daniel, S. and Bates, F.S.. "A pseudo-surfactant chemical permeation enhancer to treat otitis media via sustained transtympanic delivery of antibiotics." *Advanced Healthcare Materials*, 13, 22, 2400457 **(2024)**. ([//doi.org/10.1002/adhm.202400457](https://doi.org/10.1002/adhm.202400457))
8. Selivanovitch, E., Ostwalt, A., **Chao, Z.**, and Daniel, S.. "Emerging Designs and Applications for Biomembrane Biosensors." *Annual Review of Analytical Chemistry*, 17, 339-366 **(2024)**. ([//doi.org/10.1146/annurev-anchem-061622-042618](https://doi.org/10.1146/annurev-anchem-061622-042618))
9. Bint E Naser, S.F., Mohamed, Z.J., **Chao, Z.**, Bali, K., Owens, R.M. and Daniel, S.. "Gram-Positive Bacterial Membrane-Based Biosensor for Multimodal Investigation of Membrane-Antibiotic Interactions." *Biosensors*, 14, 45 **(2024)**. ([//doi.org/10.3390/bios14010045](https://doi.org/10.3390/bios14010045))
10. Bint E Naser, S.F., Su, H., Liu, H.Y., Manzer, Z.A., **Chao, Z.**, Roy, A., Pappa, A.M., Salleo, A., Owens, R.M. and Daniel, S.. "Detection of Ganglioside-Specific Toxin Binding with Biomembrane-Based Bioelectronic Sensors." *ACS Applied Bio Materials*, 4, 7942-7950 **(2021)**. ([//doi.org/10.1021/acsabm.1c00878](https://doi.org/10.1021/acsabm.1c00878))
11. **Chao, Z.**, Sezginel, K.B., Xu, K., Crouch, G.M., Gray, A.E., Wilmer, C.E., Bohn, P.W., Go, D.B. and Fullerton-Shirey, S.K.. "Silver Nanofilament Formation Dynamics in a Polymer-Ionic Liquid Thin Film by Direct Write." *Advanced Functional Materials*, 30, 1907950 **(2020)**. ([//doi.org/10.1002/adfm.201907950](https://doi.org/10.1002/adfm.201907950))
12. Xu, K., Liang, J., Woepfel, A., Bostian, M.E., Ding, H., **Chao, Z.**, McKone, J.R., Beckman, E.J. and Fullerton-Shirey, S.K.. "Electric Double-Layer Gating of Two-Dimensional Field-Effect

Transistors Using a Single-Ion Conductor." *ACS Applied Materials & Interfaces*, 11, 35879-35887 (2019). ([//doi.org/10.1021/acsami.9b11526](https://doi.org/10.1021/acsami.9b11526))

13. Han, D., Crouch, G.M., **Chao, Z.**, Fullerton-Shirey, S.K., Go, D.B. and Bohn, P.W.. "Nanopore-Templated Silver Nanoparticle Arrays Photopolymerized in Zero-Mode Waveguides." *Frontiers in Chemistry*, 7, 216, (2019). ([//doi.org/10.3389/fchem.2019.00216](https://doi.org/10.3389/fchem.2019.00216))
14. **Chao, Z.**, Radka, B.P., Xu, K., Crouch, G.M., Han, D., Go, D.B., Bohn, P.W. and Fullerton-Shirey, S.K.. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites." *Small*, 14, 1802023 (2018). ([//doi.org/10.1002/sml.201802023](https://doi.org/10.1002/sml.201802023))
15. Guan, J., Berlinger, S.A., Li, X., **Chao, Z.**, e Silva, V.S., Banta, S. and West, A.C.. "Development of reactor configurations for an electrofuels platform utilizing genetically modified iron oxidizing bacteria for the reduction of CO₂ to biochemicals." *Journal of Biotechnology*, 245, 21-27, (2017). ([//doi.org/10.1016/j.jbiotec.2017.02.004](https://doi.org/10.1016/j.jbiotec.2017.02.004))

CONFERENCE PRESENTATIONS

1. **Chao, Z.**; Buathong, P.; Selivanovitch, E.; Daniel, S.; Frazier, P. "Synergizing Evolution-Aware ML Model and Bioelectronics to Screen Viral Variants of Concern" 2025 AIChE 2025 Annual Meeting, Oral Presentation, Nov 2025, Boston, MA
2. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Recreating the Biological Steps of Viral Infection on a Cell-Free Bioelectronic Platform to Profile Viral Variants of Concern" 2024 CNF Annual Meeting, *Invited Talk*, Sep 2024, Ithaca, NY
3. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Decorating Bioelectronic Surface with Supported Lipid Bilayer to Study Viral Entry and More" MRS Fall 2023 Exhibition, Oral Presentation, Nov 2023, Boston, MA
4. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Electrical Sensing of SARS-CoV2 Entries Using Supported Lipid Bilayers" Asilomar Bioelectronics Symposium 2022, Poster Presentation, Sep 2022, Monterey, CA
5. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Electrical Sensing of SARS-CoV2 Entries Using Supported Lipid Bilayers" 2022 Bioanalytical Sensors Gordon Research Conference/ Seminar, Poster and Oral Presentation, Jun 2022, Newport, RI
6. **Chao, Z.**; Xu, K.; Fullerton-Shirey, S.K. "Ionic Liquid Regulated Silver Filament Formation in a Solid Polymer Electrolyte for Neuromorphic Applications" 2020 Virtual MRS Spring/Fall Meeting & Exhibit, Oral Presentation, Dec 2020
7. **Chao, Z.**; Fullerton-Shirey, S.K. "Formation/Dissolution of Silver Filaments Through an Ionic Liquid-Polymer Electrolyte Composite" MRS Fall Meeting & Exhibit, Oral Presentation, Dec 2019, Boston, MA
8. **Chao, Z.**; Gray, A.E.; Fullerton-Shirey, S.K. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites" Pittsburgh Quantum Institute 2019 Annual Conference, Poster Presentation, Apr 2019, Pittsburgh, PA
9. **Chao, Z.**; Crouch, G.; Han, D.; Bohn, P.; Go, D.; Fullerton-Shirey, S.K. "Formation/Dissolution of Silver Filaments Through an Ionic Liquid-Polymer Electrolyte Composite" AIChE Annual Meeting, Poster Presentation, Oct 2018, Pittsburgh, PA
10. **Chao, Z.**; Fullerton-Shirey, S.K. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites" Pittsburgh Quantum Institute at Pitt's Science 2018 Conference, Poster Presentation, Oct 2018, Pittsburgh, PA
11. **Chao, Z.**; Radka, B. P.; Fullerton-Shirey, S.K. "Formation/Dissolution of Conductive Silver Filaments through an Ionic Liquid/Polymer Electrolyte Thin Film" Pittsburgh Quantum Institute at Pitt's Science 2017 Conference, Poster Presentation, Oct 2017, Pittsburgh, PA

12. **Chao, Z.**; Crouch, G.; Han, D.; Bohn, P.; Go, G.; Fullerton-Shirey, S.K. "Formation/dissolution dynamics of conductive silver filaments through an ionic liquid/ polymer electrolyte thin film" 231st ECS Meeting, Oral Presentation, May 2017, New Orleans, LA

AWARDS AND FELLOWSHIPS

Schmidt AI in Science, Cornell University, Ithaca, NY <i>The Eric and Wendy Schmidt AI in Science Postdoc Fellow</i>	09/2023 to 09/2025
Smith School of Chemical and Biomolecular Engineering, Cornell University, Ithaca, NY <i>Smith Fellowship for Postdoctoral Innovation (\$ 10,000 in Research Grant)</i>	08/2022
Pittsburgh Quantum Institute, PA, USA <i>PQI 2019 Best Poster Award (\$ 1,000 in Travel Funding)</i>	05/2019
Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA <i>Outstanding Publication Award (\$ 500 in Cash Prize)</i>	12/2018
PPG Industries, Inc., PA, USA <i>Graduate Student Fellowship (\$ 25,000)</i>	03/2018
Celanese Chemical Industry Co., Ltd., China <i>Outstanding Engineer Award</i>	01/2013
Shanghai Pujing Chemical Industry Co. Ltd., China <i>Most Improved Student Award of Fly Bursary</i>	04/2012
College of Chemistry and Chemical Engineering, Nanjing Technology University, China <i>Outstanding Student Cadre</i>	12/2012
<i>Excellent Students in Three Fields</i>	12/2011
<i>Individual, second-class, third-class and first-class scholarship</i>	09/2010, 01/2011, 09/2011 and 09/2012

TEACHING EXPERIENCE

Smith School of Chemical and Biomolecular Engineering, Cornell University, NY, USA <i>Guest Speaker on Departmental Work-in-Progress Seminar</i>	10/2024
Department of Chemical and Petroleum Engineering, University of Pittsburgh, Pittsburgh, PA <i>Teaching Assistant for Undergraduate Transport Phenomenon</i>	09/2017 to 12/2017, 09/2018 to 12/2018

OUTREACH ACTIVITIES

Cornell CBE Women Events 2022, Ithaca, NY	04/2022
Volunteered the outreach activity organized by CBE WOMEN aiming at promoting interests and understandings in chemical engineering among local high school girls	
Materials Science & Technology (MS&T) Outreach, Pittsburgh, PA	10/2017
Designed and performed demo experiment for visiting middle school students near Pittsburgh area to observe polymer chain crystallization using iPad camera	